

EXPERIMENT-12

IMPLEMENTATION OF ECC-ELLIPTIC CURVE CRYPTOGRAPHY

AIM

To implement Elliptic Curve Cryptography (ECC) for encryption and decryption using elliptic curve operations.

ALGORITHM

1. Choose an elliptic curve and a base point.
2. Select a private key for the user.
3. Calculate the corresponding public key using elliptic curve point multiplication.
4. Use the public and private keys for secure encryption/decryption.
5. Perform elliptic curve point operations.
6. Display the generated keys and resulting values.

CODE:

```
main.c
1  #include <stdio.h>
2  int main()
3  {
4      int p, a, b, x, y, k;
5      printf("Enter prime p: ");
6      scanf("%d", &p);
7      printf("Enter a and b: ");
8      scanf("%d%d", &a, &b);
9      printf("Enter base point (x,y): ");
10     scanf("%d%d", &x, &y);
11     printf("Enter private key: ");
12     scanf("%d", &k);
13     printf("\nElliptic Curve: y^2 = x^3 + %dx + %d (mod %d)",
14           a, b, p);
15     printf("\nPublic Key = (%d,%d)", (k*x)%p, (k*y)%p);
16     printf("\nPrivate Key = %d", k);
17     return 0;
18 }
```

OUTPUT:

Output

```
Enter prime p: 17
Enter a and b: 2 2
Enter base point (x,y): 5 1
Enter private key: 3

Elliptic Curve:  $y^2 = x^3 + 2x + 2 \pmod{17}$ 
Public Key = (15,3)
Private Key = 3

=== Code Execution Successful ===
```

RESULT:

Thus, the basic Elliptic Curve Cryptography (ECC) concept was successfully demonstrated using elliptic curve parameters and key generation.